Flow – Workflow Builder

1. Project Overview

Build a workflow automation platform where users can:

- Create workflows in Draft
- Add trigger + action nodes
- Connect nodes in order
- Add if/else branching
- Publish workflows
- Run workflows using:
  - Webhook
  - Schedule (cron)
  - Manual Run
- View run history, logs, and failures

2. Major Functional Requirements

Workflow Management

- Create workflow
- Edit workflow in Draft
- Publish workflow
- Pause / Resume workflow
- Delete workflow
- View workflow details
- View execution history

Triggers

- Webhook Trigger
- Schedule Trigger (cron)
- Manual Trigger (Run Now button)

Node Types

- HTTP Request Node → call external API (GET/POST)
- Condition Node → true/false branching
- Delay Node → wait for N seconds
- Notify Node → send Slack or Email

Execution

- Run nodes in sequence
- Support condition branching

- Save workflow run status
- Save node-level logs
- Retry failed steps
- Show failure reason

Monitoring

- Show all runs
- Show success/failure status
- Show logs
- Show dashboard status

Authentication

- User login/register
- JWT authentication
- Basic multi-user support

3. Major Non-Functional Requirements

- Scalable → handle multiple workflow runs
- Reliable → retry failed steps
- Extensible → easy to add new node types later
- Secure → JWT auth + webhook secret
- Idempotent → avoid duplicate webhook runs
- Observable → logs, run history, health checks
- Maintainable → clean modular code

4. Main UI Screens

- Login / Register
- Dashboard
- Workflow List
- Workflow Builder Page
- Workflow Details
- Run History
- Run Details / Logs

7. Main APIs

Auth APIs

- Register - POST /api/auth/register
- Login - POST /api/auth/login

Workflow APIs

- Create Workflow - POST /api/workflows
- Get All Workflows - GET /api/workflows
- Get Workflow By Id - GET /api/workflows/{id}
- Update Workflow - PUT /api/workflows/{id}
- Delete Workflow - DELETE /api/workflows/{id}
- Publish Workflow - POST /api/workflows/{id}/publish
- Pause Workflow - POST /api/workflows/{id}/pause
- Resume Workflow - POST /api/workflows/{id}/resume

Node APIs

- Get Workflow Graph - GET /api/workflows/{id}/graph
- Add Node - POST /api/workflows/{id}/nodes
- Update Node - PUT /api/workflows/{id}/nodes/{nodeId}
- Delete Node - DELETE /api/workflows/{id}/nodes/{nodeId}

Edge APIs

- Add Edge - POST /api/workflows/{id}/edges
- Delete Edge - DELETE /api/workflows/{id}/edges/{edgeId}

Trigger APIs

- Create Webhook Trigger - POST /api/workflows/{id}/triggers/webhook
- Create Schedule Trigger - POST /api/workflows/{id}/triggers/schedule

Execution APIs

- Run Workflow Manually - POST /api/workflows/{id}/run
- Webhook Trigger API - POST /api/hooks/{webhookPath}
- Get Workflow Runs - GET /api/workflows/{id}/runs
- Get Run Details - GET /api/runs/{runId}
- Get Node Logs - GET /api/runs/{runId}/logs
- Dashboard API - GET /api/monitoring/dashboard

Database Schema

1. users

- id
- name
- email
- password_hash

- role
- created_at

2. workflows

- id
- user_id
- name
- description
- status (DRAFT, PUBLISHED, PAUSED)
- created_at
- updated_at

3. workflow_nodes

- id
- workflow_id
- name
- node_type (WEBHOOK_TRIGGER, SCHEDULE_TRIGGER, HTTP_REQUEST, CONDITION, DELAY, NOTIFY)
- config_json
- position_x
- position_y
- is_start_node

4.workflow_edges

- id
- workflow_id
- source_node_id
- target_node_id
- edge_type (DEFAULT, TRUE, FALSE)

5. workflow_triggers

- id
- workflow_id
- trigger_type (WEBHOOK, SCHEDULE)
- webhook_path
- webhook_secret
- cron_expression
- is_enabled

6. workflow_runs

- id
- workflow_id
- trigger_type
- status (QUEUED, RUNNING, SUCCESS, FAILED)
- input_payload
- output_payload
- started_at
- ended_at
- error_message

7.node_runs

- id
- workflow_run_id
- node_id
- status
- input_payload
- output_payload
- started_at
- ended_at
- error_message

8.execution_logs

- id
- workflow_run_id
- node_run_id
- log_level
- message
- created_at